PROTON-QA 1.0: A Nucleus for Provenance-Aware Ontology-Grounded Nuclear QA Benchmark

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Motivation

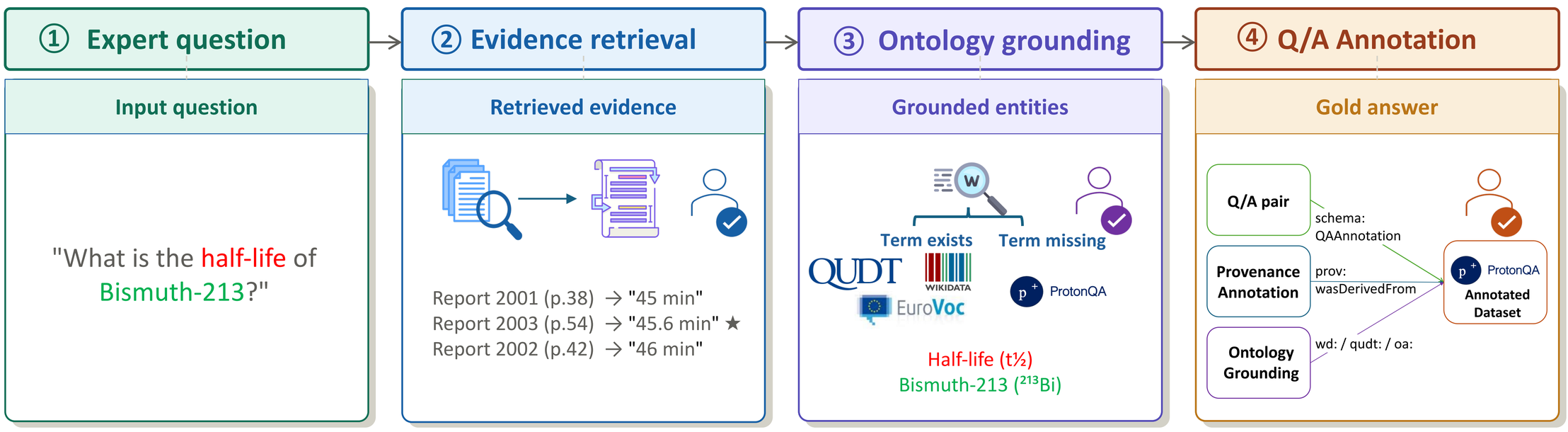
- RAG systems return plain-text answers with no provenance or domain semantics.
- Technical domains need verifiable evidence, unit-aware values and precise terminology.
- Existing QA benchmarks assume evidence is pre-identified and lack semantic grounding.

Contribution

A provenance-aware, ontology-grounded QA benchmark in RDF, providing:

- Provenance Mapping
- Unit-aware numerical answers
- Entity linking to Wikidata, EuroVoc and custom SKOS concepts
- Ontology-grounded RDF Q/A representation

Annotation Workflow



Our Expert-validated grounding workflow from expert question to semantically annotated QA pair.

RDF Model

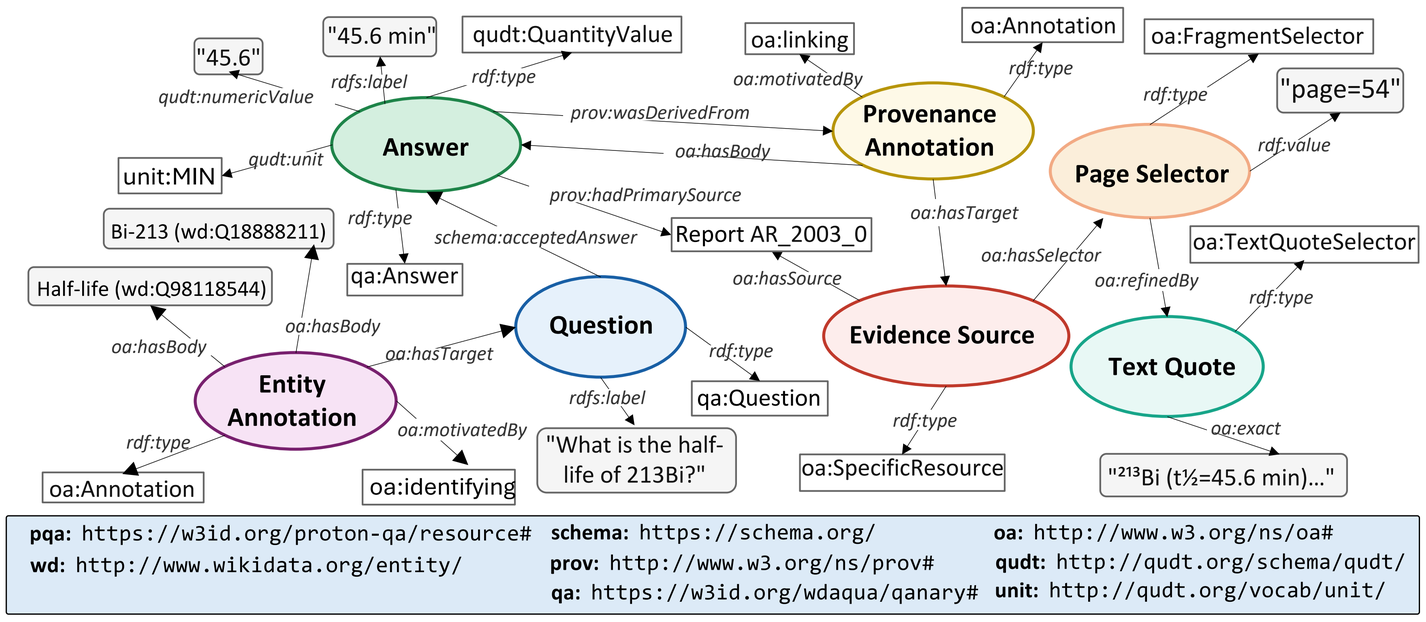


Figure 2: PROTON-QA RDF model illustrating provenance and ontology grounding.

ProtonQA Benchmark

50 expert questions

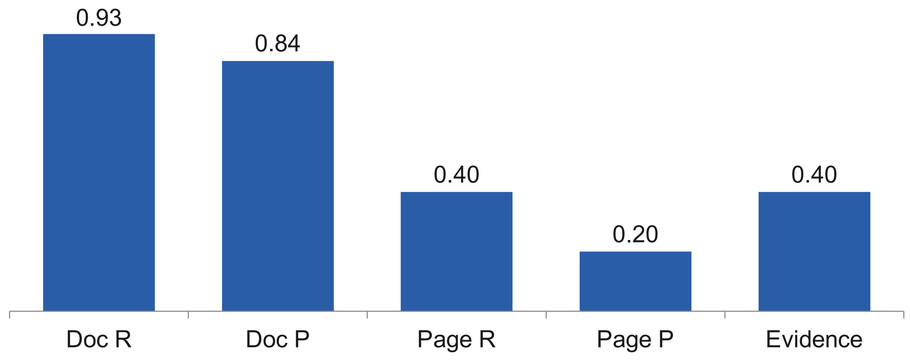
12 annual reports

14 years (2000–2013)

~1000 pages

Preliminary Evaluation

Zero-shot provenance retrieval experiment with NotebookLM



$$DocR: \frac{|R \cap G|}{|G|} \quad DocP: \frac{|R \cap G|}{|R|} \quad PageR: \frac{|R \cap G|}{|G|} \quad PageP: \frac{|R \cap G|}{|R|}$$

$$Evidence: \frac{\text{Exact span matches}}{\text{QA pairs}}$$

R: retrieved items; G: ground-truth items.

Summary:

Document-level retrieval is strong (R=0.93, P=0.84), but fine-grained provenance remains challenging. Page recall (0.40), page precision (0.20), and exact evidence matching (0.40) reveal a document-to-evidence localization gap. Despite this, answer value and unit matching reach 0.70 and 0.80, respectively, indicating that correct answers can often be recovered even when their precise provenance is not.

Comparison with Existing QA Resources

Table 1:Comparison of QA resources across semantic-web and scientific knowledge representation dimensions.

Dataset	Domain	Knowledge Source	#Qs	Representation	Ontology Grounding	Retrieval Scope	Provenance	Unit-Aware	Expert-Validated
SQuAD [6]	Open-domain	Wikipedia	107,785	JSON	None	Single-doc	Paragraph	No	No
TriviaQA [7]	Open-domain	Web & Wikipedia	95,956	JSON	None	Multi-doc	Supporting documents	No	No
QALD-9-plus [8]	Open-domain	DBpedia, Wikidata	558*	JSON	DBpedia, Wikidata	KG	KG triple(s)¶	Limited	Partial§
ACORD [9]	Legal	Legal documents	114†	JSONL/TSV	None	Single-doc	Document	No	Yes
PubMedQA [10]	Biomedical	Abstracts	1,000	JSON	None	Single-doc	Abstract	No	Yes
SPIQA [11]	Scientific	Research articles	270,000	JSON	None	Single-doc	Text, figures & tables	No	Partial§
MDA-QA [12]	Scientific	Research articles	6,804	JSON	None	Multi-doc	Supporting articles	No	Partial
PeerQA [13]	Scientific	Research articles	579	JSON	None	Single-doc	Evidence spans	No	Yes
SciDQA [14]	Computer Sci.	Research articles + reviews	2,937	JSON	None	Single-doc[a]	Text + reviews + responses	No	Yes
M3SciQA [15]	Computer Sci.	Research articles	1,452	JSON	None	Multi-doc	Paper cluster	No	Yes
NuclearQA [16]‡	Nuclear	None	100	CSV	None	None	None	No	Yes
NQuAD [17]‡	Nuclear	Technical reports	612	JSON	None	Single-doc	Passage	No	Yes
PROTON-QA	**Nuclear**	**Technical reports**	**50**	**RDF**	**Wikidata, QUDT, EuroVoc, custom**	**Corpus**	**Page + text span**	**Yes**	**Yes**

*507/558 questions include valid Wikidata queries; all include DBpedia queries.
¶SPARQL queries and returned bindings serve as provenance. §Manual, non-expert validation. †114 queries, 126,659 query–clause judgments. [a]Approximately 11% require multi-document reasoning. ‡Available only as an arXiv preprint.

Next steps larger corpus and vocabulary, multimodal evidence (tables, figures), automated annotation, and evaluation of ontology grounding across QA systems.

Data & code: github.com/ARMADA-DN/Proton-QA

Corpus: doi.org/10.5281/zenodo.22543594

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